

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 2

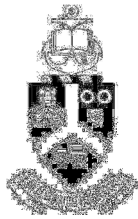
Kopiereg voorbehou

Semester Test 2

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Module EBN111
Elektrisiteit en Elektronika
8 Mei 2013

Module EBN111
Electricity and Electronics
8 May 2013



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	50	Volpunte: <i>Full marks:</i>	50
Duur van vraestel: <i>Duration of paper:</i>	90 minute <i>90 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		17	

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert
	2 JD le Roux

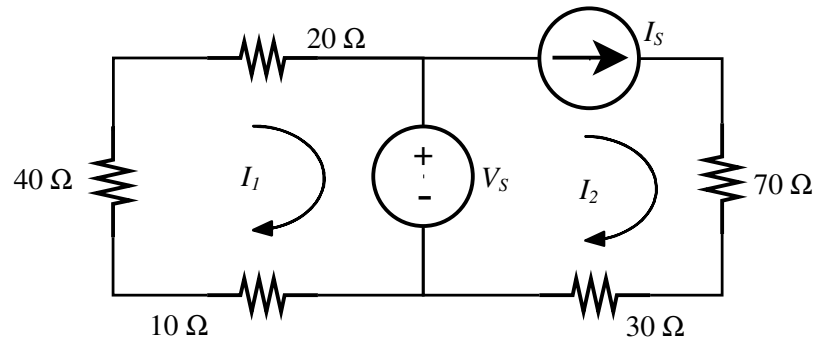
VRAAG/QUESTION 1 [25]

Vraag 1.1 [2]

Indien $I_1 = -0.25$ A en $I_2 = 0.125$ A, bepaal V_S en I_S .

Question 1.1 [2]

If $I_1 = -0.25$ A and $I_2 = 0.125$ A, determine V_S and I_S .



Vraag 1.2 [4]

Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:

$$aI_1 + bI_2 = -15$$

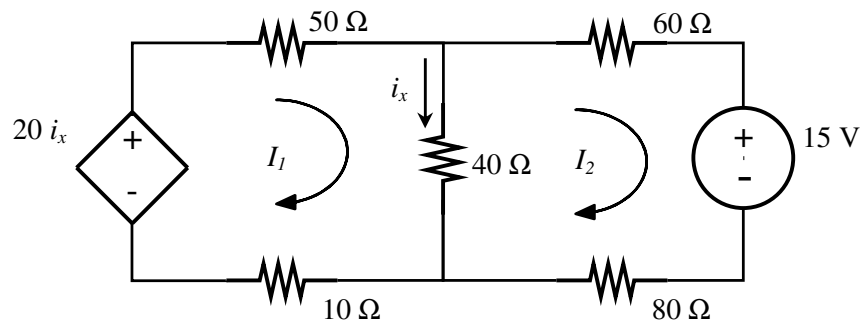
$$cI_1 + dI_2 = 0$$

Question 1.2 [4]

Use mesh analysis to set up two equations of the following form:

$$aI_1 + bI_2 = -15$$

$$cI_1 + dI_2 = 0$$

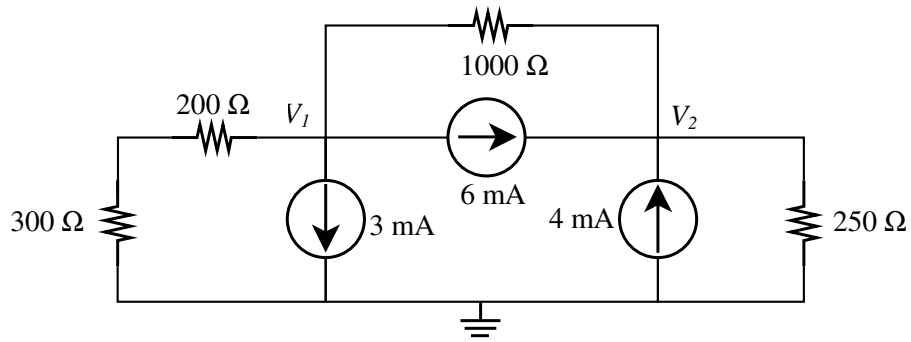


Vraag 1.3 [3]

Stel 'n matriksvergelyking d.m.v. inspeksie op in terme van die node-spannings.

Question 1.3 [3]

Use inspection to set up a matrix equation in terms of the node voltages.

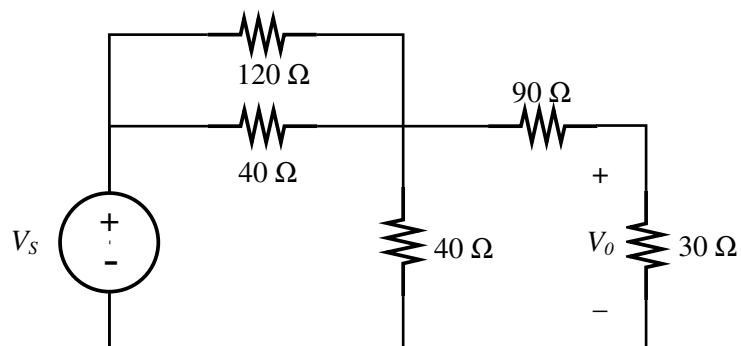


Vraag 1.4 [2]

Met behulp van die beginsel van lineariteit bepaal die verhouding V_o/V_s .

Question 1.4 [2]

Using the principle of linearity determine the relationship V_o/V_s .



Vraag 1.5 [4]

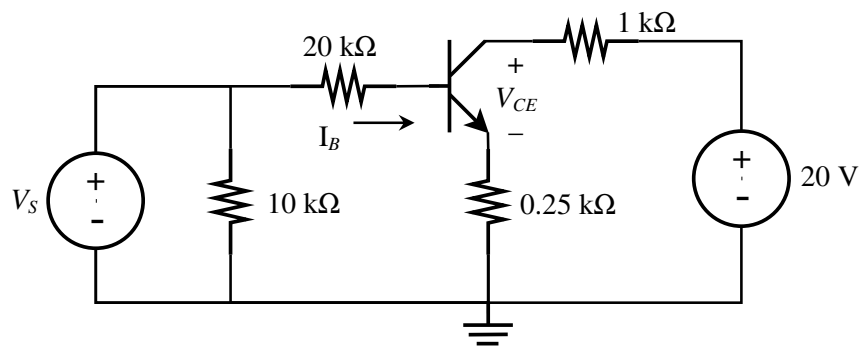
Gegee: $I_B = 200 \mu\text{A}$; $\beta = 99$. Bepaal:

- a) V_S
- b) V_{CE}

Question 1.5 [4]

Given: $I_B = 200 \mu\text{A}$; $\beta = 99$. Determine:

- a) V_S
- b) V_{CE}



Vraag 1.6 [4]

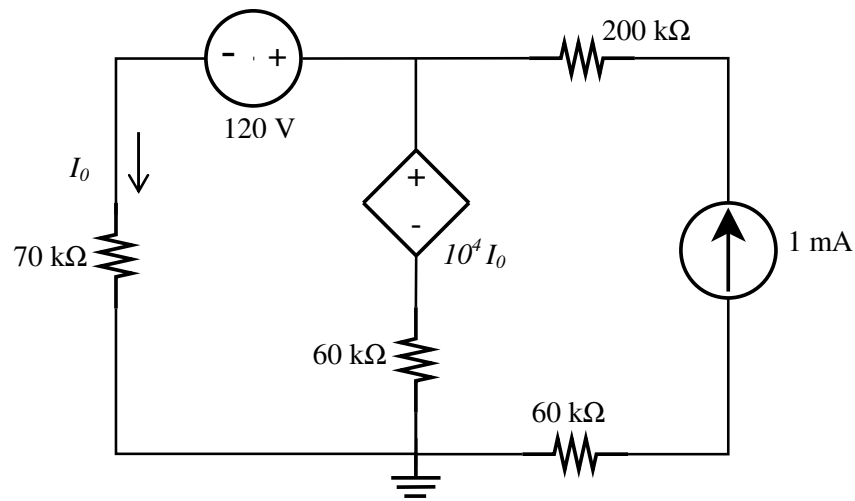
Maak gebruik van superpositie en bepaal de bydrae van:

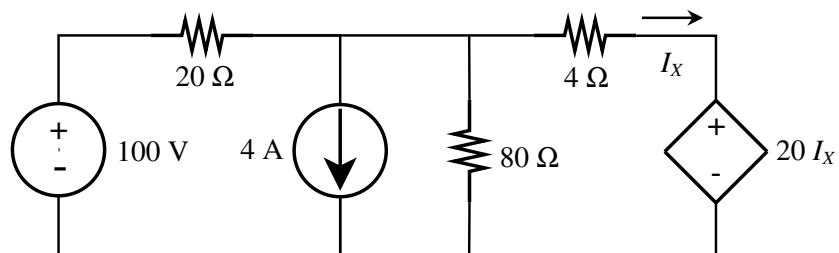
- a) die 120 V bron tot I_0 .
- b) die 1 mA bron tot I_0 .

Question 1.6 [4]

Using superposition determine the contribution of:

- a) the 120 V source to I_0 .
- b) the 1 mA source to I_0 .



Vraag 1.7 [2]Gebruik brontransformatie om I_X te bepaal.**Question 1.7 [2]**Use source transformation to determine I_X .

Vraag 1.8 [2]

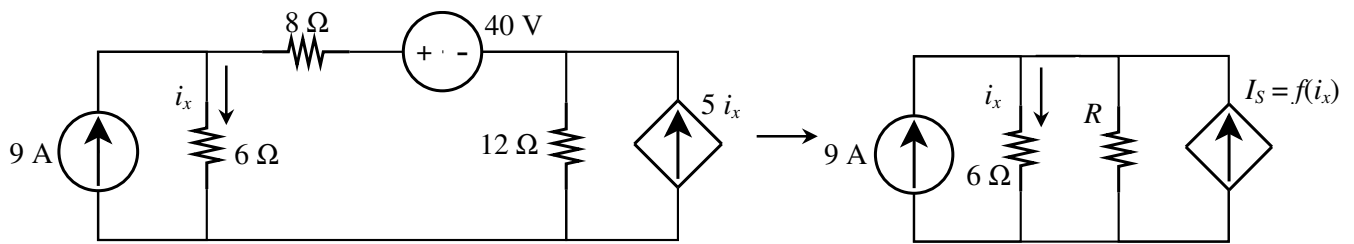
Vereenvoudig die kringbaan aan die linkerkant met brontransformasie sodat dit lyk soos die kringbaan aan die regterkant. Bepaal:

- a) R
- b) I_S as 'n funksie van i_x .

Question 1.8 [2]

Simplify the circuit on the left with source transformation so that it looks like the circuit on the right. Determine:

- a) R
- b) I_S as a function of i_x .



Vraag 1.9 [2]

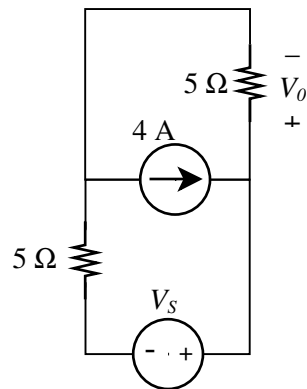
Die bydrae van V_S tot V_θ is 5 V. ($V_{0_{V_S}} = 5$ V)

- Bepaal V_S .
- Bepaal V_θ as gevolg van albei bronne.

Question 1.9 [2]

The contribution of V_S to V_θ is 5 V. ($V_{0_{V_S}} = 5$ V)

- Determine V_S .
- Determine V_θ as a result of both sources.



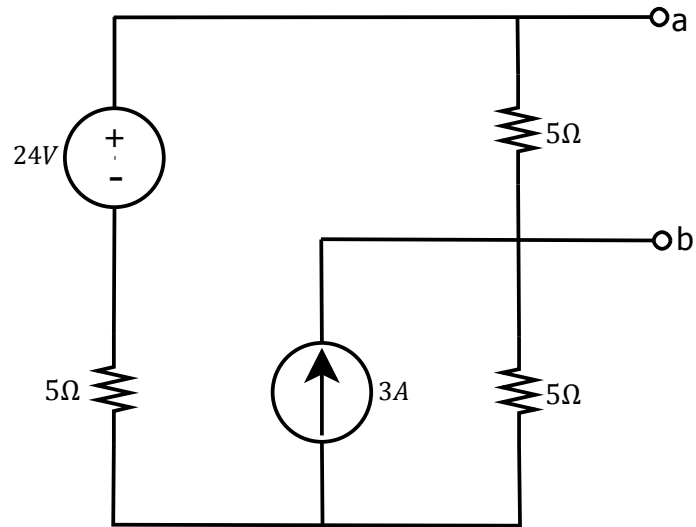
VRAAG/QUESTION 2 [25]

Vraag 2.1 [4]

Bepaal die Thevenin ekwivalente stroombaan tov terminale a-b.

Question 2.1 [4]

Determine the Thevenin equivalent circuit wrt terminals a-b.

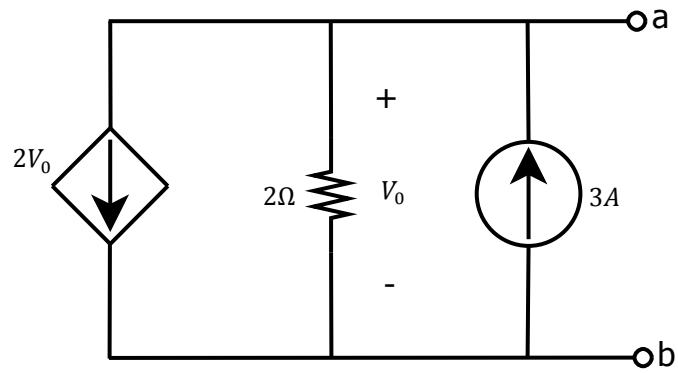


Vraag 2.2 [4]

Bepaal die Norton ekwivalente stroombaan tov terminale a-b.

Question 2.2 [4]

Determine the Norton equivalent circuit wrt terminals a-b.

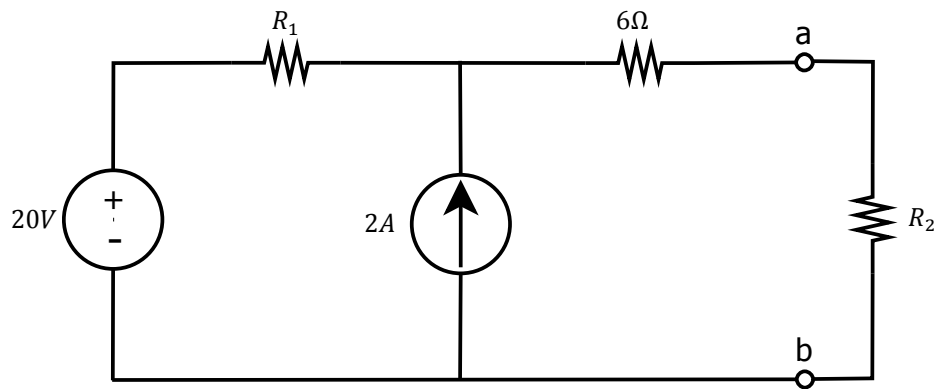


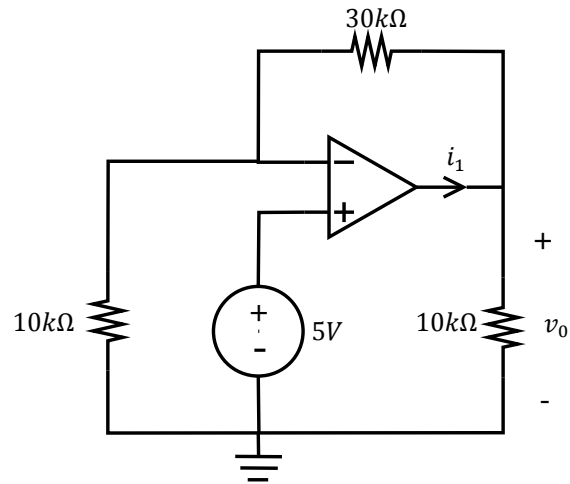
Vraag 2.3 [3]

- a) Aanvaar $R_1 = 4\Omega$, en bereken 'n waarde van R_2 sodat maksimum drywing aan R_2 oorgedra sal word.
- b) Aanvaar $R_1 = 4\Omega$, en bereken V_{Th} vir die stroombaan links van terminale a-b.
- c) Aanvaar $R_2 = 20\Omega$, en bereken 'n waarde van R_1 sodat maksimum drywing aan R_2 oorgedra sal word.

Question 2.3 [3]

- a) Assume $R_1 = 4\Omega$, and determine a value R_2 for maximum power transfer to R_2 .
- b) Assume $R_1 = 4\Omega$, and determine V_{Th} for the circuit left of terminals a-b.
- c) Assume $R_2 = 20\Omega$, and determine a value R_1 for maximum power transfer to R_2 .



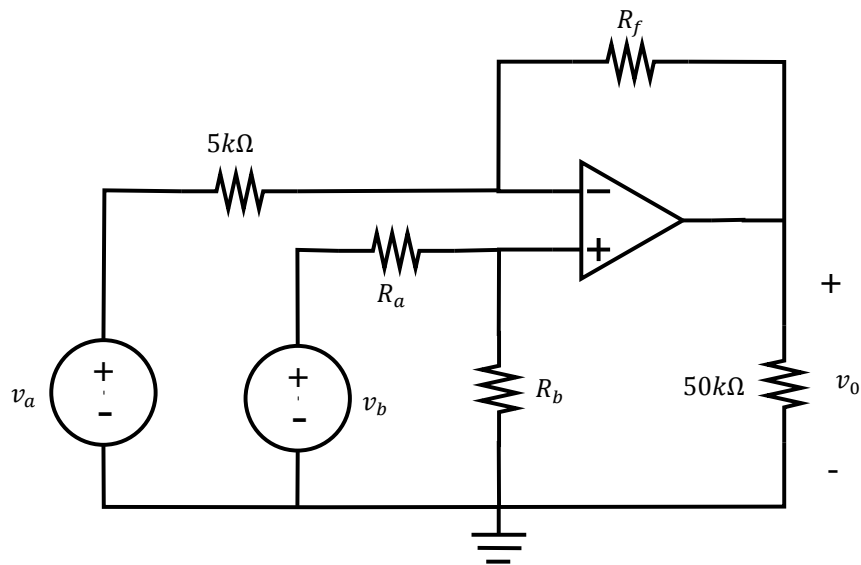
Vraag 2.4 [3]Bepaal v^- , v_o en i_I . $(v^-$ is die spanning by die negatiewe inset van die OV)**Question 2.4 [3]**Determine v^- , v_o and i_I . $(v^-$ is the voltage at the negative input of the op-amp)

Vraag 2.5 [4]

Ontwerp 'n verskil-versterker sodat $v_o = 10(v_b - v_a)$, en die spanningsbron v_b 'n intree-weerstand van $220\text{k}\Omega$ sien. Spesifiseer R_f , R_a en R_b , en bereken v^- as 'n funksie van v_a en/of v_b .

Question 2.5 [4]

Design a difference amplifier so that $v_o = 10(v_b - v_a)$, and the voltage source v_b sees an input resistance of $220\text{k}\Omega$. Specify R_f , R_a and R_b , and calculate v^- as a function of v_a and/or v_b .

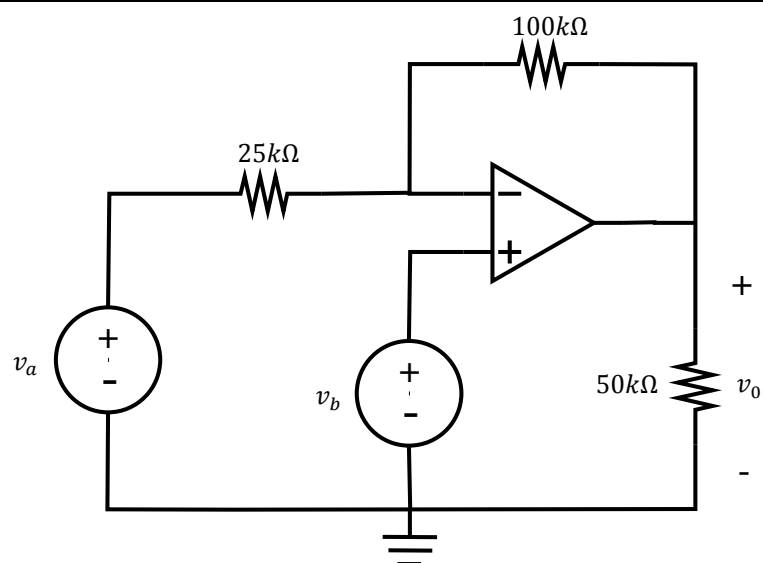


Vraag 2.6 [3]

- a) Bereken v_o as $v_a = 4\text{V}$ en $v_b = 0\text{V}$.
- b) Bereken v_o as $v_a = 2\text{V}$ en $v_b = 1\text{V}$.
- c) Bereken v_o as $v_a = 0\text{V}$ en $v_b = 2\text{V}$.

Question 2.6 [3]

- a) Determine v_o if $v_a = 4\text{V}$ and $v_b = 0\text{V}$.
- b) Determine v_o if $v_a = 2\text{V}$ and $v_b = 1\text{V}$.
- c) Determine v_o if $v_a = 0\text{V}$ and $v_b = 2\text{V}$.



Vraag 2.7 [4]

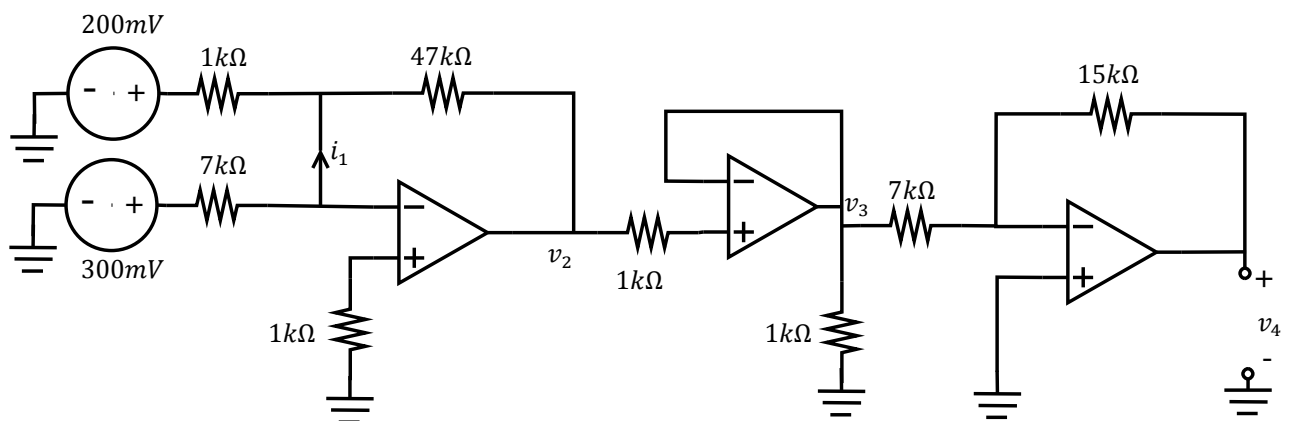
Bereken

- a) i_1
- b) v_2
- c) v_3/v_2
- d) v_4/v_3

Question 2.7 [4]

Determine

- a) i_1
- b) v_2
- c) v_3/v_2
- d) v_4/v_3



EBN111 SEMESTER TOETS/TEST 2 (8 MEI/MAY 2013)
ANTWOORDBLAD / ANSWER SHEET

Van / Surname:		Studentennummer: Student number:							
Voorletters / Initials:		Tel. nr. / Tel no.:							
Dissipline / Discipline:		Taal / Language:				Afr	Eng		

Q1		Q2		Tot:	/50
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VRAAG 1 – QUESTION 1 (25)		
1.1	$V_S =$	$I_S =$
1.2	$a =$	$b =$
	$c =$	$d =$
1.3	$\begin{bmatrix} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} \end{bmatrix}$	
1.4	$V_0 / V_S =$	
1.5	$a) V_S =$	$b) V_{CE} =$
1.6	$a) I_{O(120V)} =$	$b) I_{O(1mA)} =$
1.7	$I_X =$	
1.8	$a) R =$	$b) I_S = f(i_x) =$
1.9	$a) V_S =$	$b) V_0 =$

VRAAG 2 – QUESTION 2 (25)

2.1	$V_{TH} =$		$R_{TH} =$
2.2	$I_N =$		$R_N =$
2.3	$R_2 =$	$V_{Th} =$	$R_l =$
2.4	$v^- =$	$v_0 =$	$i_l =$
2.5	$R_f =$		$v^- =$
	$R_a =$		$R_b =$
2.6	$a) \ v_0 =$	$b) \ v_0 =$	$c) \ v_0 =$
2.7	$i_l =$		$v_2 =$
	$v_3/v_2 =$		$v_4/v_3 =$